



Clim1001quiz3 - practice

Introduction to Climate Change (University of New South Wales)



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What does it mean when a tree's rings grow wider than the previous year?

Select one:

- ☐ a. The tree received less rainfall/ the temperature was cooler during the growing season that year
- ☒ b. The tree received more rainfall/ the temperature was warmer during the growing season that year ✓
- ☐ c. Nothing - this is unrelated to climate

Your answer is correct.

The correct answer is: The tree received more rainfall/ the temperature was warmer during the growing season that year

Why do we use climate projections? Choose the **best** answer:

Select one:

- ☐ a. They help us understand the economic impact of emissions reduction strategies
- ☐ b. They allow us to estimate our 'carbon budget'
- ☐ c. They allow us to view the climate impacts of different emissions scenarios
- ☐ d. They help to understand the social cost of climate change
- ☐ e. They provide us with possible impacts due to future warming
- ☒ f. all of the above ✗
- ☐ g. a and b only
- ☐ h. b, c and e only
- ☐ i. c, d and e only

Your answer is incorrect.

The correct answer is: b, c and e only

Different isotopes of an element:

Select one:

- ☒ a. have different masses and a different number of neutrons ✓
- ☐ b. have different masses and a different number of protons
- ☐ c. have the same mass and the same number of protons
- ☐ d. have the same mass and the same number of neutrons

The correct answer is: have different masses and a different number of neutrons

Global sea level rise over the last 50 years is due to:

Select one:

- ☐ a. the sea getting warmer
- ☐ b. sea ice melting
- ☐ c. land ice melting
- ☐ d. plate tectonics
- ☒ e. all of the above ✗
- ☐ f. a and c only
- ☐ g. none of the above

The correct answer is: a and c only

Approximately what proportion of global energy consumption is sourced from fossil fuels?

Select one:

- ☐ a. 50%
- ☐ b. 30%
- ☒ c. 85% ✓
- ☐ d. 99%

The correct answer is: 85%

What percentage of heat from global warming has the ocean absorbed in the past 40 years?

Select one:

- ☐ a. less than 10%
- ☐ b. about 50%
- ☐ c. about 20%
- ☒ d. more than 80% ✓

What is a paleoclimate proxy? Choose the best answer:

Select one:

- ☐ a. Any climate record that relies on chemical isotopes
- ☐ b. Tree rings, coral and sediments but not historical documents
- ☒ c. An indirect measure of climate prior to instrumental records ✓
- ☐ d. Any record of the climate prior to the industrial revolution

Your answer is correct.

The correct answer is: An indirect measure of climate prior to instrumental records

Emissions scenarios and Representative Concentration Pathways (RCPs) are used to:

Select one:

- ☒ a. inform climate model projections about future changes in greenhouse gas concentrations ✓
- ☐ b. explain the sources of past greenhouse gas emissions
- ☐ c. constrain different countries' greenhouse gas emissions
- ☐ d. ensure there is no uncertainty due to greenhouse gas emissions levels into the future

The correct answer is: inform climate model projections about future changes in greenhouse gas concentrations

Generally speaking, models are

Select one:

- ☐ a. too simplistic to improve our understanding of what they represent
- ☒ b. an approximation of some aspect of the real world ✓
- ☐ c. perfect and highly reliable for making predictions
- ☐ d. an exact representation of some aspect of the real world

The correct answer is: an approximation of some aspect of the real world

The energy efficiency of an incandescent light bulb is around:

Select one:

- ☐ a. 50%
- ☐ b. 3%
- ☒ c. 10% ✓
- ☐ d. 30%

The correct answers are: 3%, 10%, 30%, 50%

What is the dominant feedback in initiating a glacial period?

Select one:

- ☐ a. Milankovitch cycles
- ☒ b. The ice-albedo feedback ✓
- ☐ c. Lack of winter insolation at 65°N
- ☐ d. Carbon cycle feedbacks

Your answer is correct.

The correct answer is: The ice-albedo feedback

What is the purpose of the COPs?

Select one:

- ☒ a. To review the Convention, amend it and develop new climate change law ✓
- ☐ b. To help implement mitigation and adaptation strategies in developing countries
- ☐ c. To deal with administrative tasks such as organising conferences
- ☐ d. To aid the conference of the parties in scientific and technical matters

Which of the following checks to see that the goals of the [Paris Agreement](#) are being met?

Select one:

- ☐ a. The Conference of the Parties
- ☐ b. NDCs
- ☐ c. The Compliance Committee
- ☒ d. The Global Stocktake ✓

Your answer is correct.

The correct answer is: The Global Stocktake

Which of the following is correct?

Select one:

- ☐ a. Annex 1 countries include developing countries but not transitioning economies
- ☐ b. Non-Annex countries refer to developed countries with non-transitioning economies.
- ☒ c. Annex 1 countries include transitioning economies but not developing countries ✓
- ☐ d. Annex 2 countries are those with transitioning economies

Which of the following is **not** one of the three main elements of the Kyoto Protocol?

Select one:

- ☐ a. Reporting and review
- ☐ b. Flexible mechanisms
- ☒ c. Joint implementation ✓
- ☐ d. Compliance

Your answer is correct.

The correct answer is: Joint implementation

What proportion of the energy in coal ends up as electricity produced by a coal-fired power station?

Select one:

- ☒ a. 30-40% ✓
- ☐ b. 3-5%
- ☐ c. 10-15%
- ☐ d. 50-70%

An example of a Kyoto protocol "flexibility mechanism" is:

Select one:

- ☐ a. carbon taxation
- ☐ b. common but differentiated responsibilities
- ☒ c. joint implementation ✓
- ☐ d. intergenerational equity

The correct answer is: joint implementation

What is the maximum temperature goal for global warming under the Paris Agreement?

Select one:

- ☐ a. 1°C
- ☒ b. 2°C ✓
- ☐ c. 1.5°C
- ☐ d. 3.5°C

Melting sea ice could cause global sea level to rise by as much as 6 meters...

Select one:

- ☒ a. False ✓
- ☐ b. True

Your answer is correct.

The correct answer is: False

What do we mean by the term 'paleoclimate'? Choose the best answer:

Select one:

- ☒ a. Climates older than those recorded directly with instruments ✓
- ☐ b. Climate that is measured using tree ring records
- ☐ c. The climate during the period when the dinosaurs were around
- ☐ d. The climate of the Paleocene 66-56 million years ago

Why is 'technical status' an important consideration when evaluating an energy generation technology? Choose the best answer:

Select one:

- ☐ a. if it's too technical it will be hard to implement effectively
- ☐ b. we have no idea if it actually works unless it has been commercialised
- ☐ c. older technologies are generally not very popular
- ☒ d. the costs, potential scale, and speed of deployment are not yet known if the technology is immature ✓

The correct answer is: the costs, potential scale, and speed of deployment are not yet known if the technology is immature

Increasing the resolution of a global climate model (for example from 100 to 50km **grid cell** widths) means that (choose the best answer)

Select one:

- ☐ a. the computational cost of running the model increases
- ☐ b. the computational cost of running the model decreases
- ☐ c. the **time step** size of the model has to decrease
- ☐ d. the **time step** size of the model has to increase
- ☐ e. a and c
- ☒ f. a and d ✗
- ☐ g. b and d

Your answer is incorrect.

The correct answer is: a and c

Choose the **best** answer: current state of the art climate models:

Select one:

- ☐ a. solve the physical equations governing the [climate system](#)
- ☐ b. are tested against observations
- ☐ c. couple the atmosphere and ocean only
- ☐ d. simulate land surface processes such as [photosynthesis](#) and transpiration
- ☐ e. simulate the global economy to estimate future greenhouse gas emissions
- ☐ f. all of the above
- ☒ g. a, b and d only ✓
- ☐ h. a and b only
- ☐ i. a, b and c only

Your answer is correct.

The correct answer is: a, b and d only

In current [UNFCCC](#) jargon, 'NDC's are ...

Select one:

- ☐ a. Nationally Detrended Cuts
- ☐ b. Neutral Density Carbon units
- ☐ c. The carbon budget of each signatory to the Kyoto Protocol
- ☐ d. Developing nations in the [UNFCCC](#)
- ☒ e. The contribution each nation is willing to make to achieve the aims of the [Paris agreement](#) ✓
- ☐ f. none of these

During the last few decades, sea levels around the globe have:

Select one:

- ☐ mostly decreased, but increased in some places
- ☒ mostly increased, but decreased in some places ✓
- ☐ decreased everywhere
- ☐ increased everywhere

The correct answer is: mostly increased, but decreased in some places

If all the ice on Greenland and Antarctica were to melt, by how much would global sea levels rise?

Select one:

- ☐ a. about 2 metres
- ☐ b. about 10 metres
- ☒ c. about 70 metres ✓
- ☐ d. about half a metre
- ☐ e. about 1 metre

What is meant by the 'energy trilemma'?

Select one:

- ☐ a. The trade-off between cost, speed of deployment and environmental benefits
- ☐ b. The choice between energy frugality, efficiency and supply changes
- ☒ c. The trade-off between between energy security, equity and environmental sustainability ✓
- ☐ d. The choice between coal, nuclear or renewable energy sources

Your answer is correct.

The correct answer is: The trade-off between between energy security, equity and environmental sustainability

The Paleocene-Eocene thermal maximum (PETM) is mainly of interest to climate scientists because:

Select one:

- ☐ a. It shows that anthropogenic global warming will lead to mass extinction
- ☐ b. it is an example of a glaciation period not caused by orbital changes
- ☒ c. it is an example of rapid warming, like today ✓
- ☐ d. it might explain why the dinosaurs became extinct
- ☐ e. many climate records disagree about its existence

Global climate model simulations are typically performed at a coarse resolution (e.g. 100km x 100m **grid cells**) because:

Select one:

- ☒ a. Our understanding of the physics of the **climate system** at smaller scales is limited ✖
- ☐ b. We only have large scale observations to validate models with
- ☐ c. Supercomputers do not provide enough computational power to run models at higher resolutions
- ☐ d. They are the only spatial scales at which models are good at reproducing temperature trends of the last century

Your answer is incorrect.

The correct answer is: Supercomputers do not provide enough computational power to run models at higher resolutions

The last glacial period ended around:

Select one:

- ☐ a. 100,000 years ago
- ☐ b. 1,000,000 years ago
- ☐ c. 500 years ago
- ☒ d. 10,000 years ago ✔

What is the relevance of Maslow's pyramid to energy markets?

Select one:

- ☐ a. Access to commercial energy has historically been the least important of several factors
- ☒ b. Energy security is generally viewed as more important than environmental concerns associated with energy production ✓
- ☐ c. Debate around energy policy is difficult when it comes economic efficiency and energy security
- ☐ d. The social acceptability of energy sources tends to dominate decision making

Your answer is correct.

The correct answer is: Energy security is generally viewed as more important than environmental concerns associated with energy production

Which of the following are key factors when deciding on new electricity generation?

Select one:

- ☐ a. Speed and scale of deployment
- ☐ b. Present and future costs
- ☐ c. Delivered environmental benefits
- ☒ d. All of the above ✓
- ☐ e. a and b only

What determines the acidity of a liquid?

Select one:

- ☐ a. The concentration of CO₂ in the liquid
- ☒ b. The concentration of hydrogen ions in the liquid ✓
- ☐ c. The concentration of carbonate ions in the liquid
- ☐ d. The concentration of iron in the liquid

Your answer is correct.

The correct answer is: The concentration of hydrogen ions in the liquid

The Convention outlines the principles that are to guide international action on climate change. Which of the following is **not** one of those principles?

Select one:

- ☐ a. Sustainable development
- ☐ b. Intergenerational equity
- ☒ c. Wealth redistribution ✓
- ☐ d. Precautionary principle

The current global mean sea level is the highest we have observed in Earth's history...

Select one:

- ☒ a. False ✓
- ☐ b. True

Your answer is correct.

The correct answer is: False

Which of the following can result (at least in part) from climate change?

Select one:

- ☐ a. Crop failures
- ☐ b. Armed conflict
- ☐ c. Expansion and settlement of new lands
- ☒ d. All of the above ✓

Your answer is correct.

The correct answer is: All of the above

Why should we be concerned about ocean acidification?

Select one:

- ☐ a. The increased acidity will make it unsafe to swim in the ocean
- ☐ b. It will increase coastal erosion
- ☒ c. It impacts the base of the ocean food web ✓
- ☐ d. It threatens to change major ocean currents

Your answer is correct.

The correct answer is: It impacts the base of the ocean food web

Which of the following represents the fastest, cheapest available option for immediate reduction in carbon emissions from the electricity sector?

Select one:

- ☐ a. nuclear power
- ☒ b. energy efficiency ✓
- ☐ c. solar PV
- ☐ d. carbon capture and storage

What is meant by 'detection and attribution' ?

Select one:

- ☐ a. Finding out where climate change is happening
- ☐ b. The process of finding out which countries are most responsible for greenhouse gas emissions
- ☐ c. Identifying natural event cycles El Nino
- ☒ d. Determining whether an event is more likely due to anthropogenic climate change or natural variability ✓

Your answer is correct.

The correct answer is: Determining whether an event is more likely due to anthropogenic climate change or natural variability

Current levels of greenhouse gas emissions and atmospheric concentrations are:

Select one:

- ☒ a. significantly higher than those used in RCPs used to drive climate models shown in the IPCC 5th assessment report ✓
- ☐ b. significantly lower than those used in RCPs used to drive climate models shown in the IPCC 5th assessment report
- ☐ c. similar to those used in RCPs used to drive climate models shown in the IPCC 5th assessment report